

Module 6: Lesson 4

Portfolio choice with a sparse autoencoder



Outline

- ▶ Building and training a sparse autoencoder.
- ▶ Autoencoder-based portfolio choice.

Building a sparse autoencoder

In this lesson, we build and train a sparse autoencoder to extract the latent factors that explain the variation of the 20 tech companies' stock returns that we already analyzed in Lesson 2.

We use the autoencoder model to design a portfolio that optimizes the risk-return trade-off as measured by the Sharpe Ratio.

The model is designed as an overcomplete linear autoencoder with a single hidden layer, whose dimension is equal to the number of stocks, $k = n = 20$.

The model shrinks the dimensionality of the encoding vector by including a Lasso-type regularization term in the loss function:

$$\left\| X - \hat{X} \right\|_F^2 + \lambda_1 \left\| \text{vec}(Z) \right\|_1$$

The value of the hyperparameter λ_1 determines how narrow is the bottleneck of the autoencoder.

Remember that we can implement the sparsity term in TensorFlow by specifying an activity regularizer:

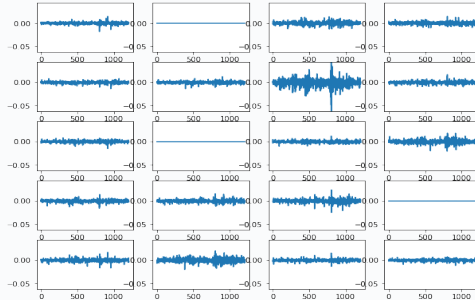
```
activity_regularizer= tf.keras.regularizers.l1(l1weight)
```

Training a sparse autoencoder

We train the autoencoder using the data from the 20 stocks, with the usual 80-20 training-test sample split.

We set aside an additional 20% of the training sample left for validation using an early stopping criterion.

- ▶ We train the model for 1,000 epochs, setting $\lambda_1 = 0.0025$.
- ▶ Training leaves 17 encodings as “relevant,” generating a small reconstruction error of roughly 5%.



Autoencoder-based portfolio choice: A model for stock returns

The trained model can be used as a basis for the design of investment strategies.

- ▶ As a byproduct, the autoencoder generates estimates for the expected returns of each stock, the exposures to systematic factors, and the amount of noise.
- ▶ We can use all this information to design diversified portfolios.

The decoding layer of the autoencoder implies that we can model the returns of stocks at time t as:

$$R_t = \hat{R}_t + \mathbf{e}_t = \mathbf{a} + B\mathbf{F}_t + \mathbf{e}_t$$

where

- ▶ $\mathbf{a}$ is an $n \times 1$ vector that includes all the trained biases in the output layer.
- ▶ B is an $n \times k$ matrix that includes all the trained weights of the output layer.
- ▶ $\mathbf{F}_t$ is a $k \times 1$ vector of realizations of the factors, which in a single-layer model are just the encodings, $\mathbf{Z}_t$.
- ▶ $\mathbf{e}_t$ is an $n \times 1$ vector of reconstruction errors, that we assume of zero-mean and uncorrelated.

Autoencoder-based portfolio choice: Risk and expected return

Using the model for stock returns from the autoencoder, we can recover estimates for the expected returns (μ) and the variance-covariance matrix of returns (Σ) from the training sample.

- ▶ $\mu = E(\hat{R}_t) = \mathbf{a} + BE(F_t)$: This is the average of the reconstructions in the training sample.
- ▶ $\Sigma = \text{Var}(R_t) = \text{Var}(\hat{R}_t) + \text{Var}(\mathbf{e}_t)$: This is the variance-covariance matrix of returns of the reconstructions, generated by the factors and the exposures, plus the diagonal (by assumption) variance-covariance matrix of reconstruction errors.

Let a portfolio be defined by a vector $w = (w_1, \dots, w_n)^T$ of weights, such that $\sum_i^n w_i = 1$.

The expected return and variance of a portfolio w can be expressed as:

$$\begin{aligned} E(R_t^w) &= w^T \mu \\ \text{Var}(R_t^w) &= w^T \Sigma w \end{aligned}$$

And the Sharpe Ratio is given by $SR^w = \frac{w^T \mu}{(w^T \Sigma w)^{1/2}}$.

Autoencoder-based portfolio choice: Performance

We devise an optimization problem, whereby we want to choose a portfolio that maximizes the return per unit of risk, as measured by the Sharpe Ratio, given the estimated μ and σ from the autoencoder.

- ▶ The solution is analytical if we do not impose short-selling restrictions.
- ▶ The optimal portfolio yields an estimated, annualized, Sharpe Ratio of 2.89 in the training sample.

We evaluate the performance of the resulting portfolio in the test sample (**Disclaimer:** These results refer to a particular initialization of the trainable parameters and may vary if other seeds are used).

- ▶ The out-of-sample Sharpe Ratio is just 0.09.
- ▶ Portfolios based on other strategies (using the sample averages and variance-covariance matrix or PCA) yield substantial negative Sharpe ratios.

The test sample features a quite different market environment than the training sample, which naturally leads to generalization issues.

However, the autoencoder seems to extract informative features that significantly outperform other, more naïve, estimation methods.

Summary of Lesson 4

In Lesson 4, we have:

- ▶ Built and trained a sparse autoencoder to extract the latent factors from a sample of stock returns.
- ▶ Used the trained model to design a portfolio that maximizes the Sharpe Ratio.

⇒ **References for this lesson:**

Conlon, T. et al. "Machine Learning and Factor-Based Portfolio Optimization." *arXiv*, 2021.

TO DO NEXT: Please go to the Jupyter Notebook associated with Lesson 4.

In the next module, we will describe transformer methods.

See you there!